

AEROFIT

Treadmill Customer Analysis

Detailed Exploratory Data Analysis & Customer Profiling Report

Project Type	Exploratory Data Analysis
Dataset	Aerofit Treadmill Customer Data
Records	180 customers
Products	KP281 • KP481 • KP781
Technology	Python • Pandas • NumPy • Matplotlib • Seaborn

Prepared as a stakeholder-oriented summary of the supplied Google Colab analysis.

1. Executive Summary

This project analyzes customer characteristics associated with AeroFit's three treadmill models—KP281, KP481, and KP781. The objective is to understand how demographic, financial, usage, and fitness characteristics differ across products and to build practical customer profiles that can support product positioning and recommendations.

The supplied analysis contains 180 observations and 9 variables. The dataset is clean: no missing values and no duplicate records were identified. The analysis combines descriptive statistics, categorical distributions, histograms, distribution plots, count plots, product-level boxplots, and correlation analysis.

The results show clear differences in the customer characteristics associated with the three products. KP281 has the largest share of customers, while KP781 is associated with customers showing higher income, usage, fitness, and expected weekly miles. The correlation analysis also indicates strong positive relationships between usage, fitness, and expected weekly miles.

Important interpretation: these findings describe patterns in the supplied customer dataset. They should be treated as associations rather than evidence that one characteristic causes a customer to purchase a particular treadmill.

2. Business Problem & Objectives

AeroFit offers multiple treadmill models designed for different customer needs. A key business question is whether customer characteristics can be used to distinguish the profiles associated with each product.

The analysis therefore focuses on the following objectives:

- Understand the structure and quality of the customer dataset.
- Describe the demographic and behavioural characteristics of customers.
- Compare customer characteristics across KP281, KP481, and KP781.
- Identify relationships among income, usage, fitness, and expected weekly miles.
- Translate analytical observations into practical customer-profile and marketing implications.

3. Dataset Overview

Attribute	Description
Rows	180 customer records
Columns	9 variables
Product	KP281, KP481, KP781
Age	Customer age in years
Gender	Male / Female
Education	Years of education
Marital Status	Single / Partnered
Usage	Average number of treadmill sessions planned per week
Fitness	Self-rated fitness on a 1–5 scale

Attribute	Description
Income	Annual income in dollars
Miles	Expected weekly miles walked/run

The notebook reports that the dataset contains 6 numerical and 3 categorical variables. All 180 records are complete, with no missing values or duplicate records. Categorical fields were converted to the category datatype during preparation. [filecite turn3file0 L21-L38](#) [filecite turn3file0 L43-L62](#)

4. Descriptive Statistics

Variable	Mean	Std. Dev.	Min	Median	Max
Age	28.79	6.94	18	26	50
Education	15.57	1.62	12	16	21
Usage	3.46	1.08	2	3	7
Fitness	3.31	0.96	1	3	5
Income	\$53,719.58	\$16,506.68	\$29,562	\$50,596.50	\$104,581
Miles	103.19	51.86	21	94	360

- Customer age ranges from 18 to 50 years, with a mean of approximately 28.8 years.
- Weekly treadmill usage ranges from 2 to 7 sessions, with an average of about 3.46 sessions.
- Self-rated fitness ranges from 1 to 5, with an average of 3.31.
- Annual income varies substantially, from \$29,562 to \$104,581.
- Expected weekly miles range from 21 to 360, indicating substantial variation in expected activity levels.

These statistics indicate a heterogeneous customer base with meaningful variation in financial capacity and fitness/activity expectations. [filecite turn3file0 L63-L85](#)

5. Customer & Product Distribution

Dimension	Category	Count	Share
Product	KP281	80	44.44%
Product	KP481	60	33.33%
Product	KP781	40	22.22%
Gender	Male	104	57.78%
Gender	Female	76	42.22%
Marital Status	Partnered	107	59.44%
Marital Status	Single	73	40.56%

KP281 is the most represented product in the sample, followed by KP481 and KP781. The customer base is moderately male-skewed and has a higher proportion of partnered customers. These distributions describe the

sample and should not automatically be interpreted as market-wide shares. filecite turn3file0 L137-L180

6. Exploratory Data Analysis

The notebook uses several complementary visualization techniques to understand the distributions and product-level differences.

6.1 Univariate Analysis

- Histograms and distribution plots were generated for Age, Education, Usage, Fitness, Income, and Miles.
- Age is concentrated around the younger adult portion of the sample, with a right tail toward older customers.
- Income has a broad distribution with higher-income observations extending beyond the main concentration.
- Miles is positively skewed, with a smaller number of customers reporting substantially higher expected mileage.

6.2 Categorical Analysis

- Count plots were used for Education, Usage, and Fitness.
- Usage is concentrated around 2–4 weekly sessions, with fewer observations at the highest usage levels.
- Fitness is concentrated around the middle of the 1–5 scale, while both lower and higher fitness levels are represented.

6.3 Product-Level Analysis

- Boxplots compare Age, Income, Miles, Usage, and Fitness across the three products.
- The product comparisons show increasingly distinct customer characteristics for the higher-end KP781 model, particularly for income, expected miles, usage, and fitness.

The notebook's product boxplots provide the primary evidence for these profile differences; exact product-level medians were not tabulated in the supplied analysis, so this report does not invent additional numerical values.

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7. Product Customer Profiles

Product	Observed profile from analysis	Potential business use
KP281	Largest customer group in the dataset. Customer ages and income levels are generally lower than the KP781 group, and they are more likely to be used for casual or recreational purposes.	Position as the budget-friendly option that the KP781 group actively characterizes.
KP481	Middle product group by customer count. Its customer characteristics are generally between the lower and higher-end products, with a mix of usage and fitness levels.	Acts as a bridge product between the lower and higher-end products, offering a middle ground in terms of price and performance.
KP781	Smaller customer group but associated with higher income, usage, and expected miles.	Targeted at serious fitness enthusiasts and professionals who require high-performance treadmills.

The profile descriptions above are based on the product-level boxplots and the observations recorded in the notebook. They represent patterns within this 180-record dataset, not a universal definition of AeroFit customers.

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8. Correlation Analysis

Relationship	Correlation	Interpretation
Usage ↔ Miles	0.759	Strong positive association
Fitness ↔ Miles	0.786	Strong positive association

Relationship	Correlation	Interpretation
Usage ↔ Fitness	0.669	Moderate-to-strong positive association
Education ↔ Income	0.626	Moderate-to-strong positive association
Income ↔ Miles	0.543	Moderate positive association
Income ↔ Fitness	0.535	Moderate positive association
Age ↔ Income	0.513	Moderate positive association

The strongest relationship in the supplied correlation matrix is between Fitness and Miles (0.786), followed by Usage and Miles (0.759). Usage and Fitness also have a positive relationship of 0.669. This suggests that customers expecting greater activity also tend to report higher fitness levels and higher intended usage.

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Statistical caution: correlation measures association, not causation. The results do not establish that increasing fitness causes higher treadmill usage or that income causes customers to purchase a particular product.

9. Business Recommendations

Based on the observed patterns, AeroFit can use the analysis to develop a more structured customer-profiling approach.

- **Segment by activity intensity:** Use expected weekly miles, usage, and fitness level as practical indicators when recommending treadmill models.
- **Consider income alongside activity needs:** Income shows a meaningful positive association with fitness, usage, and miles, and the KP781 profile is associated with higher income in the product comparison.
- **Build product-specific messaging:** Marketing communication can be differentiated around broad accessibility for the largest customer group and higher-performance positioning for customers with stronger activity expectations.
- **Use customer profiling for recommendations:** A simple rules-based recommendation framework could combine income, usage, fitness, and expected miles to suggest a suitable model.
- **Validate with additional data:** Before using these profiles for major commercial decisions, test them against a larger and more representative customer population.

10. Analytical Methodology

Stage	Approach
Data Loading	Imported the AeroFit treadmill CSV dataset using Pandas.
Data Quality	Checked dataset shape, information, missing values, and duplicate records.
Data Preparation	Converted Product, Gender, and MaritalStatus to categorical variables.
Descriptive Analysis	Calculated counts, percentages, central tendency, dispersion, and ranges.
Univariate EDA	Used histograms and distribution plots to inspect numerical variables.
Categorical EDA	Used count plots for discrete variables.
Product Comparison	Used boxplots to compare customer characteristics across treadmill models.
Relationship Analysis	Calculated a Pearson correlation matrix and visualized it as a heatmap.

The supplied notebook uses Pandas, NumPy, Matplotlib, and Seaborn for the analysis and visualization workflow.

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11. Limitations

- The dataset contains 180 observations and may not represent AeroFit's full customer population.
- The analysis is descriptive and exploratory; it does not establish causal relationships.
- The dataset does not provide transaction dates, so time-series, cohort, retention, or trend analysis cannot be performed from the supplied data.
- The analysis does not include product price, profit margin, acquisition cost, or marketing-channel data, limiting direct profitability analysis.
- Expected miles and usage are customer-level activity expectations and should not be interpreted as verified actual treadmill usage.
- The correlation matrix describes linear association among numerical variables and does not account for all possible confounding factors.

12. Future Scope

- Perform statistical hypothesis testing to validate whether product-level differences are statistically significant.
- Create customer segments using clustering or business-rule segmentation.
- Build a product recommendation model based on customer characteristics.
- Use logistic regression or other classification methods to model product selection.
- Add transaction history to enable customer lifetime value, retention, cohort, and time-series analysis.
- Build an interactive Power BI dashboard for stakeholder exploration and product-level decision support.
- Evaluate marketing effectiveness by combining customer profiles with campaign and conversion data.

13. Conclusion

The AeroFit customer analysis demonstrates how exploratory data analysis can convert a structured customer dataset into useful product and customer profiles. The 180-record dataset is clean and contains sufficient variation to identify meaningful differences across the three treadmill models.

KP281 represents the largest portion of the sample, while KP781 is associated with a more active and higher-income customer profile. The correlation analysis further shows strong positive associations among expected miles, usage, and fitness.

For business use, the most practical next step is to combine these descriptive profiles with statistical testing, a larger customer sample, and additional commercial data. This would allow AeroFit to move from descriptive profiling toward validated segmentation and recommendation strategies.

Appendix A — Key Dataset Values

Metric	Value
Total observations	180
Total variables	9
Product categories	3
Male customers	104 (57.78%)
Female customers	76 (42.22%)
Partnered customers	107 (59.44%)
Single customers	73 (40.56%)
KP281 customers	80 (44.44%)
KP481 customers	60 (33.33%)
KP781 customers	40 (22.22%)
Average age	28.79 years
Average education	15.57 years
Average usage	3.46 sessions/week
Average fitness	3.31 / 5
Average income	\$53,719.58
Average expected miles	103.19 miles/week

Source: Supplied Aerofit Google Colab analysis PDF. All numerical findings in this report are derived from the supplied analysis; where the source did not provide exact product-level summary values, the report uses qualitative descriptions rather than estimating values.

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